

From the Dirichlet Integral to Lobachevsky’s Formula: A Formalization in Lean 4

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Abstract

We formalize the Dirichlet integral and several of its classical applications in the Lean 4 proof assistant. Since the sinc function is not Lebesgue integrable on the positive half-line, the Dirichlet integral must be represented as the limit of integrals over bounded intervals. To avoid the difficulty of removing an exponential factor from a conditionally convergent integral, we instead pass through the absolutely integrable function sinc^2 . We evaluate its integral by differentiation under the integral sign and dominated convergence, and then recover the Dirichlet integral from an identity between truncated integrals. Using these results, we formalize the convergence of the Dirichlet cutoff to the Heaviside function and derive several quadratic and bilinear trigonometric integral identities. Finally, we formalize Lobachevsky’s integral formula for continuous periodic functions satisfying a reflection symmetry, using the density of cosine polynomials obtained from Mathlib’s Fourier analysis on the additive circle.

1 Introduction

The Dirichlet integral is a classical example of a conditionally convergent improper integral. Although its statement involves only elementary functions, its evaluation already exhibits several ideas that are central to real and Fourier analysis.

1.1 The Dirichlet integral and its historical origins

The Dirichlet integral is the identity

$$\lim_{T \rightarrow +\infty} \int_0^T \frac{\sin t}{t} dt = \frac{\pi}{2}. \quad (1)$$

Throughout the paper, we use the continuous extension

$$\text{sinc}(t) = \begin{cases} \sin(t)/t, & t \neq 0, \\ 1, & t = 0. \end{cases}$$

The integral arose from the development of Fourier analysis. Fourier’s study of the heat equation introduced representations of functions by trigonometric series [1], but their convergence was not yet rigorously understood. In his 1829 memoir, Dirichlet gave the first general convergence theorem for Fourier series under suitable regularity assumptions [2].

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By expressing the partial sums through what is now called the Dirichlet kernel, Dirichlet reduced the convergence problem to the study of oscillatory integrals. One of the central limits in his argument has the form

$$\lim_{N \rightarrow +\infty} \int_0^h f(\beta) \frac{\sin(N\beta)}{\sin \beta} d\beta = \frac{\pi}{2} f(0),$$

under appropriate continuity and monotonicity assumptions on f near the origin. For $f = 1$, a rescaling of the integration variable leads to the Dirichlet integral.

The Dirichlet integral subsequently became a standard example of a conditionally convergent improper integral [3, 4].

A concrete consequence is obtained from the normalized primitive

$$S(x) := \frac{1}{2} + \frac{1}{\pi} \int_0^x \text{sinc}(t) dt. \quad (2)$$

Equation (1) and the evenness of sinc imply that

$$\lim_{R \rightarrow +\infty} S(Rx) = \begin{cases} 0, & x < 0, \\ \frac{1}{2}, & x = 0, \\ 1, & x > 0. \end{cases}$$

Thus, the rescaled primitive converges pointwise to the Heaviside function. This illustrates the role of the Dirichlet integral in Fourier inversion and in the approximation of discontinuous functions [5].

The sinc function later became fundamental in harmonic analysis and signal processing. Up to normalization, it is the inverse Fourier transform of the indicator function of an interval and appears as the interpolation kernel in the Shannon sampling theorem [6].

Finally, the Dirichlet integral leads to further identities involving sine and sinc functions, as well as to Lobachevsky's integral formula. We formalize these results in the Lean 4 proof assistant [7], using the Mathlib library [8]. The complete formal development is available in the accompanying repository [9].

1.2 Challenges in the formalization

The main difficulty in formalizing the Dirichlet integral lies not in the evaluation of its value, but in the choice of an appropriate notion of integral. In informal mathematics, the notation

$$\int_0^{+\infty} \text{sinc}(t) dt$$

is commonly used without further qualification. Since sinc is not absolutely integrable, however, this expression cannot be interpreted as an ordinary Lebesgue integral. In Mathlib, the Bochner integral of a non-integrable function is defined to be zero. Therefore, the formal statement of the Dirichlet integral must concern the convergence of the truncated integrals:

$$\int_0^T \text{sinc}(t) dt \longrightarrow \frac{\pi}{2} \quad \text{as } T \rightarrow +\infty.$$

In Lean, this convergence is expressed using the filters `atTop` and `Tendsto`. This formulation faithfully represents conditional convergence, but it also prevents several standard results about Lebesgue integrals from being applied directly.

This distinction also affects the classical argument based on the effect of the exponential decay on the kernel. Although the integrals are absolutely convergent, removing the exponential cannot be justified by a direct application of the dominated convergence theorem: the natural pointwise majorant is $|\text{sinc}|$, which is not integrable.

Our formalization therefore follows a different proof, in which the conditionally convergent integral of sinc is recovered from the absolutely convergent integral of sinc^2 .

1.3 Main contributions

This paper presents a Lean 4 formalization of three closely related families of results in real analysis:

- **The Dirichlet integral.** We formalize the Dirichlet integral as the convergence of integrals over bounded intervals.
- **Further integral identities.** We develop several consequences of the Dirichlet integral. In particular, we formalize the convergence of the Dirichlet cutoff to the Heaviside function, together with a collection of real and complex trigonometric integral identities.
- **Lobachevsky’s integral formula.** We formalize Lobachevsky’s formula for continuous periodic functions satisfying the appropriate symmetry assumptions.

1.4 Related formalization work and structure of the paper

The Dirichlet integral was previously formalized in Isabelle/HOL by Avigad, Hölzl, and Serafin as part of their formalization of the central limit theorem [10]. Their proof represents $1/x$ as an integral of e^{-ux} , exchanges the resulting integrals using Fubini’s theorem, and applies dominated convergence after evaluating the inner integral. Our formalization follows a different route: it first evaluates the absolutely convergent integral of sinc^2 , and then recovers the Dirichlet integral from an identity between truncated integrals. In both developments, conditional convergence is expressed explicitly as a limit of integrals over bounded intervals.

To the best of our knowledge, Lobachevsky’s integral formula has not previously been formalized in a proof assistant. The corresponding part of our development therefore goes beyond a new formalization of the Dirichlet integral: it provides a machine-checked proof of a general identity for continuous periodic functions, combining the integral identities established earlier with uniform approximation by cosine polynomials.

The approximation argument builds on Mathlib’s Fourier analysis on the additive circle. This infrastructure has also been used in the formalization of zeta and L -functions by Loeffler and Stoll [11]; our contribution here is its application to Lobachevsky’s formula rather than the underlying Fourier theory.

Structure of the paper. Section 2 explains the obstruction to formalizing the classical Feynman argument directly, and presents the proof through sinc^2 together with its Lean implementation. Section 3 derives the Dirichlet cutoff and the corresponding Heaviside limit, as well as several quadratic and product integral identities. Section 4 establishes Lobachevsky’s formula, beginning with cosine polynomials and then passing to continuous periodic functions by uniform approximation. The final section discusses possible extensions of the development and concludes the paper.

2 The Dirichlet Integral: Classical Proofs and Formalization

The Dirichlet integral is one of the classical examples of an improper integral and, in many undergraduate analysis courses, the first example in which convergence is genuinely conditional. Its apparently elementary integrand conceals the essential phenomenon: the integral converges through oscillatory cancellation, although the absolute value is not integrable.

2.1 The classical Feynman argument and its formal obstruction

A familiar evaluation of the Dirichlet integral begins by introducing an exponential inverse factor. Following the method commonly associated with Feynman [12], define,

for $a > 0$,

$$F(a) = \int_0^{+\infty} e^{-at} \operatorname{sinc}(t) dt.$$

Differentiation under the integral sign is justified for every $a > 0$, and yields

$$F'(a) = - \int_0^{+\infty} e^{-at} \sin t dt = - \frac{1}{1+a^2}.$$

Since $F(a) \rightarrow 0$ as $a \rightarrow +\infty$, it follows that

$$F(a) = \int_a^{+\infty} \frac{du}{1+u^2} = \frac{\pi}{2} - \arctan(a).$$

In many textbook presentations, the proof is essentially concluded at this point by letting a tend to zero under the integral sign:

$$\begin{aligned} \lim_{a \rightarrow 0^+} \int_0^{+\infty} e^{-at} \operatorname{sinc}(t) dt &\stackrel{?}{=} \int_0^{+\infty} \lim_{a \rightarrow 0^+} e^{-at} \operatorname{sinc}(t) dt \\ &= \int_0^{+\infty} \operatorname{sinc}(t) dt. \end{aligned}$$

The conclusion is correct, but it requires an additional estimate, uniform in the exponential parameter. For $a, T > 0$, an integration by parts gives the uniform estimate

$$\left| \int_T^{+\infty} e^{-at} \operatorname{sinc}(t) dt \right| \leq \frac{3}{T}.$$

which, together with dominated convergence on each fixed interval $[0, T]$, justifies the passage $a \rightarrow 0^+$.

Thus, the argument is valid, but its usual presentation hides an additional uniform tail estimate. Formalizing it would require coordinating the limits $a \rightarrow 0^+$ and $T \rightarrow +\infty$, together with an integration-by-parts argument at an infinite endpoint.

We instead pass through sinc^2 , which is absolutely integrable on the positive half-line. The parameter can then be removed directly by dominated convergence, and an identity between truncated integrals of sinc and sinc^2 transfers the resulting value back to the Dirichlet integral.

2.2 An integrable detour through the squared sinc function

Our proof first evaluates the absolutely convergent integral of sinc^2 , and then transfers its value to the conditionally convergent Dirichlet integral. The squared sinc function is integrable on the positive half-line: it is bounded near the origin and satisfies

$$\operatorname{sinc}^2(t) \leq \frac{1}{t^2} \quad (t \geq 1).$$

The connection between the two integrals follows from a change of variables and an integration by parts. For every $T > 0$, one obtains

$$\int_0^T \operatorname{sinc}(t) dt = \int_0^{T/2} \operatorname{sinc}^2(t) dt + \frac{T}{2} \operatorname{sinc}^2\left(\frac{T}{2}\right). \quad (3)$$

The boundary term tends to zero, since

$$0 \leq \frac{T}{2} \operatorname{sinc}^2\left(\frac{T}{2}\right) \leq \frac{2}{T}.$$

It therefore remains to evaluate the integral of sinc^2 .

For $x > 0$, introduce the exponentially integral

$$I(x) = \int_0^{+\infty} e^{-xt} \operatorname{sinc}^2(t) dt.$$

Differentiating twice under the integral sign gives

$$I''(x) = \int_0^{+\infty} e^{-xt} \sin^2(t) dt = \frac{1}{2x} - \frac{x}{2(4+x^2)}.$$

Together with the behavior of I and I' at infinity, this determines

$$I(x) = \frac{x}{4} \log \left(\frac{x^2}{4+x^2} \right) + \arctan \left(\frac{2}{x} \right). \quad (4)$$

Since sinc^2 is integrable, dominated convergence allows the factor to be removed as $x \rightarrow 0^+$. Taking the limit in (4) yields

$$\int_0^{+\infty} \operatorname{sinc}^2(t) dt = \frac{\pi}{2}.$$

Finally, letting $T \rightarrow +\infty$ in (3) gives

$$\lim_{T \rightarrow +\infty} \int_0^T \operatorname{sinc}(t) dt = \frac{\pi}{2}.$$

The advantage of this route is that dominated convergence is applied only to an absolutely integrable function. Conditional convergence is handled separately through the identity (3).

2.3 The Lean formalization

The formalization keeps the two notions of integration separate. The squared sinc function is treated as a Lebesgue-integrable function on the positive half-line, whereas the Dirichlet integral is stated as the convergence of bounded interval integrals.

The first step is the integrability of the squared sinc function. The proof splits the positive half-line at 1, using continuity on the bounded part and comparison with t^{-2} at infinity.

```
1 lemma integrable_sinc_sq :
2   IntegrableOn (fun (t : ℝ) ↦ (sinc t)^2) (Ioi 0)
```

The identity (3) requires slightly more care. Since integration by parts involves $t \mapsto -1/t$, it is first proved away from the origin. The lower endpoint is then sent to zero.

```
1 lemma integral_sinc_sq_eq_dirichlet_bounded {a T : ℝ} (ha : 0 < a) (hT : a ≤ T)
2   :
3   (∫ t in a..T, Real.sinc t) = (∫ t in a / 2..T / 2, (Real.sinc t)^2)
4   - (Real.sinc (a / 2))^2 * (a / 2) + (Real.sinc (T / 2))^2 * (T /
5   2)
6 lemma integral_sinc_zero_T (T : ℝ) (hT : T > 0) :
7   (∫ t in 0..T, Real.sinc t) =
8   (∫ t in 0..T / 2, (Real.sinc t)^2)
   + (Real.sinc (T / 2))^2 * (T / 2)
```

To evaluate the squared sinc integral, the development introduces the integral I and its first two derivatives.

```

1 def sinc_sq_times_exp (t : ℝ) : ℝ → ℝ :=
2   fun x ↦ Real.exp (-x * t) * (Real.sinc t)^2
3
4 def neg_sinc_sq_times_id_exp (t : ℝ) : ℝ → ℝ :=
5   fun x ↦ -(Real.sinc t)^2 * t * Real.exp (-x * t)
6
7 def sin_sq_times_exp (t : ℝ) : ℝ → ℝ :=
8   fun x ↦ (Real.sin t)^2 * Real.exp (-x * t)
9
10 def integral_sinc_sq_times_exp (x : ℝ) : ℝ :=
11   ∫ t in Ioi 0, sinc_sq_times_exp t x
12
13 def integral_neg_sinc_sq_times_id_exp (x : ℝ) : ℝ :=
14   ∫ t in Ioi 0, neg_sinc_sq_times_id_exp t x
15
16 def integral_sin_sq_times_exp (x : ℝ) : ℝ :=
17   ∫ t in Ioi 0, sin_sq_times_exp t x

```

For a fixed $x > 0$, differentiation under the integral sign is carried out in a neighborhood where the parameter remains bounded below by $x/2$. This provides an integrable exponential majorant and yields the two derivative identities.

```

1 theorem hasDeriv_integral_sinc_sq_times_exp (x : ℝ) (hx : 0 < x) :
2   HasDerivAt (integral_sinc_sq_times_exp) (integral_neg_sinc_sq_times_id_exp x)
3   x
4 theorem hasDeriv_integral_neg_sinc_sq_times_id_exp (x : ℝ) (hx : 0 < x) :
5   HasDerivAt (integral_neg_sinc_sq_times_id_exp) (integral_sin_sq_times_exp x)
6   x

```

After evaluating the second derivative, the closed form is identified from its derivative and its limit at infinity.

```

1 theorem integral_sinc_sq_times_exp_eq (x : ℝ) (hx : 0 < x) :
2   integral_sinc_sq_times_exp x = x / 4 * Real.log (x^2 / (4 + x^2)) + Real.
3   arctan (2 / x)

```

Dominated convergence then removes the exponential factor. The resulting evaluation is combined with `integral_sinc_zero_T` to obtain the Dirichlet integral.

```

1 theorem integral_sinc_sq_eq_pi_div_two :
2   ∫ t in Ioi 0, (Real.sinc t)^2 = π / 2
3
4
5 theorem integral_dirichlet :
6   Tendsto (fun T ↦ ∫ t in 0..T, sinc t) atTop (ℕ (π / 2))

```

The final theorem therefore has the intended meaning: the Dirichlet integral is a limit of truncated integrals, not the Lebesgue integral of the non-integrable sinc function. The separate evaluation of sinc^2 will also be used in the applications below.

3 Further Integral Identities and the Dirichlet Cutoff

The Dirichlet integral becomes particularly useful after scaling its upper endpoint. We first use this observation to construct a smooth approximation of the Heaviside function and then derive a cutoff formula for integrable functions. Further trigonometric identities will be considered in the following subsections.

3.1 The Dirichlet cutoff and the Heaviside function

Define the normalized primitive

$$D(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^x \text{sinc}(t) dt$$

and the Heaviside function

$$H(x) = \begin{cases} 1, & x > 0, \\ 1/2, & x = 0, \\ 0, & x < 0. \end{cases}$$

The Dirichlet integral and the evenness of sinc imply that, for every $x \in \mathbb{R}$,

$$\lim_{R \rightarrow +\infty} D(Rx) = H(x). \quad (5)$$

The corresponding definitions and convergence theorem in Lean are:

```

1 noncomputable def DirichletSin : ℝ → ℝ :=
2   fun x ↦ 1/2 + 1/π * ∫ t in (0)..(x), sinc t
3
4
5 noncomputable def HeavisidePerso (x : ℝ) : ℝ :=
6   if x > 0 then 1 else if x = 0 then 1/2 else 0
7
8
9 theorem lim_SRx (x : ℝ) :
10  Tendsto (fun R ↦ DirichletSin (R * x)) atTop (ℕ (HeavisidePerso x))

```

The formalization also establishes the standard analytic properties of D , including its continuity and boundedness on $\mathbb{R}$.

```

1 lemma DirichletSin_continuous :
2   Continuous fun u ↦ DirichletSin (u)
3
4 theorem DirichletSinBounded :
5   ∃ M, ∀ y, |DirichletSin y| ≤ M

```

This bound allows the pointwise convergence (5) to be used under an integral. If f is integrable and $t \in \mathbb{R}$, then

$$\lim_{R \rightarrow +\infty} \int_{\mathbb{R}} f(a) D(R(a-t)) da = \int_t^{+\infty} f(a) da. \quad (6)$$

```

1 theorem Tendsto_Integral_DirichletSin_times_integrableFunction
2   (f : ℝ → ℝ) (t : ℝ) (hf : Integrable (fun t ↦ f t)) :
3   Tendsto (fun T : ℝ ↦ ∫ a, f a * DirichletSin (T * (a - t))) atTop (ℕ (∫
4     a in Ici t, f a))

```

The development also proves the complex-valued analogue `Tendsto_Integral_DirichletSin_times_integrableFunction'` and a version restricted to the positive half-line, `Tendsto_Integral_DirichletSin_times_integrableFunction_zero'`.

```

1 theorem Tendsto_Integral_DirichletSin_times_integrableFunction' (f : ℝ → ℂ) (t
2   : ℝ) (hf : Integrable (fun t ↦ f t)) :
3   Tendsto (fun T : ℝ ↦ ∫ a, f a * ↑(DirichletSin (T * (a - t)))) atTop (ℕ (
4     ∫ a in Ici t, f a))

1 theorem Tendsto_Integral_DirichletSin_times_integrableFunction_zero' (f : ℝ →
2   ℂ) (t : ℝ) (hf : Integrable (fun t ↦ f t)) :
3   Tendsto (fun T : ℝ ↦ ∫ a in Ici 0, f a * ↑(DirichletSin (T * (a - t))))
4     atTop (ℕ (∫ a in Ici (max 0 t), f a))

```

These cutoff formulas will be used in the integral identities below.

3.2 Quadratic trigonometric integrals

The evaluation of the squared sinc integral immediately yields several parameterized identities.

3.2.1 Scaling the squared sinc integral

For $a > 0$, the change of variables $u = at$ gives

$$\int_0^{+\infty} \text{sinc}^2(at) dt = \frac{\pi}{2a}. \quad (7)$$

```
1 lemma integral_sinc_sq_scaled_of_pos (a : ℝ) (ha : 0 < a) :
2   (∫ t in Ioi 0, (Real.sinc (a * t)) ^ 2) = (1 / a) * (Real.pi / 2)
```

3.2.2 The quadratic sine kernel

Since

$$\left(\frac{\sin(at)}{t}\right)^2 = a^2 \text{sinc}^2(at)$$

for $t > 0$, Equation (7), together with the symmetry in a , gives

$$\int_0^{+\infty} \left(\frac{\sin(at)}{t}\right)^2 dt = \frac{\pi|a|}{2}. \quad (8)$$

```
1 theorem integral_sin_sq_div_sq (a : ℝ) :
2   (∫ t in Ioi 0, (Real.sin (a * t) / t) ^ 2) = Real.pi * |a| / 2
```

The corresponding integrability statement is formalized as `integrable0n_sin_sq_div_sq`.

3.2.3 The cosine-difference kernel

Using

$$1 - \cos u = 2 \sin^2\left(\frac{u}{2}\right),$$

Equation (8) yields

$$\int_0^{+\infty} \frac{1 - \cos(at)}{t^2} dt = \frac{\pi|a|}{2}. \quad (9)$$

```
1 theorem integral_one_sub_cos_div_sq (a : ℝ) :
2   (∫ t in Ioi 0, (1 - Real.cos (a * t)) / t ^ 2) = Real.pi * |a| / 2
```

Its integrability is recorded in `integrable0n_one_sub_cos_div_sq`. This last form is particularly useful for evaluating products of sine functions.

3.3 Products of sine and sinc functions

The cosine-difference identity also gives closed forms for products of sine and sinc functions.

3.3.1 Products of sine functions

For $a, b \geq 0$, the product-to-sum identity gives

$$2 \sin(at) \sin(bt) = (1 - \cos((a+b)t)) - (1 - \cos((a-b)t)).$$

Applying Equation (9) yields

$$\begin{aligned} \int_0^{+\infty} \frac{\sin(at) \sin(bt)}{t^2} dt &= \frac{\pi}{4} (a + b - |a - b|) \\ &= \frac{\pi}{2} \min\{a, b\}. \end{aligned}$$

```
1 theorem integral_sin_mul_sin_div_sq (a b : ℝ) (ha : 0 ≤ a) (hb : 0 ≤ b) :
2   (∫ t in Ioi 0, Real.sin (a * t) * Real.sin (b * t) / t ^ 2) = Real.pi * min
   a b / 2
```

3.3.2 Products of scaled sinc functions

For $a, b > 0$,

$$\operatorname{sinc}(at) \operatorname{sinc}(bt) = \frac{1}{ab} \frac{\sin(at) \sin(bt)}{t^2}$$

almost everywhere on the positive half-line. Consequently,

$$\begin{aligned} \int_0^{+\infty} \operatorname{sinc}(at) \operatorname{sinc}(bt) dt &= \frac{\pi \min\{a, b\}}{2ab} \\ &= \frac{\pi}{2 \max\{a, b\}}. \end{aligned}$$

```
1 theorem integral_sinc_mul_sinc (a b : ℝ) (ha : 0 < a) (hb : 0 < b) :
2   (∫ t in Ioi 0, Real.sinc (a * t) * Real.sinc (b * t)) = Real.pi / (2 * max
   a b)
```

These identities provide the trigonometric integral evaluations used in the proof of Lobachevsky's formula.

4 Lobachevsky's Integral Formula

The identities established in the preceding section are instances of a broader phenomenon: when the squared sinc function is multiplied by a suitably symmetric periodic function, the integral over the entire positive half-line reduces to an integral over a single half-period. Lobachevsky's formula makes this principle precise.

4.1 Historical background and statement

Although Nikolai Lobachevsky is primarily remembered for non-Euclidean geometry, he also worked on probability, trigonometric series, and definite integrals. The formulas now bearing his name originate in his 1842 memoir on the mean results of repeated observations [13]. They extend the Dirichlet integral by allowing the sinc kernel to be weighted by a periodic function.

Later generalizations have considered higher powers of the sinc function [14] and interpretations based on Fourier analysis, Parseval-type identities, and the Shannon sampling basis [15]. We formalize the classical formula for the absolutely integrable squared sinc kernel.

Theorem 1 (Lobachevsky's integral formula). *Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous and satisfy*

$$f(x + \pi) = f(x) \quad \text{and} \quad f(\pi - x) = f(x)$$

for every $x \in \mathbb{R}$. Then

$$\int_0^{+\infty} \text{sinc}^2(x) f(x) dx = \int_0^{\pi/2} f(x) dx.$$

The symmetry assumptions imply that f is even and suggest approximation by cosine polynomials

$$p(x) = \sum_{n=0}^N a_n \cos(2nx).$$

The proof first establishes the formula for each cosine mode and then passes to a uniform limit.

4.2 The proof for cosine polynomials

For $n \geq 1$, the identity

$$\begin{aligned} \text{sinc}^2(x) \cos(2nx) = \frac{1}{4} & \left(\frac{1 - \cos(2(n+1)x)}{x^2} + \frac{1 - \cos(2(n-1)x)}{x^2} \right. \\ & \left. - 2 \frac{1 - \cos(2nx)}{x^2} \right) \end{aligned}$$

and Equation (9) give

$$\int_0^{+\infty} \text{sinc}^2(x) \cos(2nx) dx = 0.$$

For $n = 0$, the integral is $\pi/2$. These are also the values of the corresponding integrals over $[0, \pi/2]$:

$$\int_0^{\pi/2} \cos(2nx) dx = \begin{cases} \pi/2, & n = 0, \\ 0, & n \geq 1. \end{cases}$$

The nonconstant mode calculation is formalized as follows.

```
1 lemma integral_sinc_sq_mul_cos_two_nat (n : ℕ) (hn : 0 < n) :
2   (∫ x in Set.Ioi 0, (Real.sinc x) ^ 2 * Real.cos (2 * (n : ℝ) * x)) = 0
```

For

$$p(x) = \sum_{n=0}^N a_n \cos(2nx),$$

linearity now gives

$$\int_0^{+\infty} \text{sinc}^2(x) p(x) dx = \int_0^{\pi/2} p(x) dx. \quad (10)$$

```
1 def cosinePolynomial (N : ℕ) (a : ℕ → ℝ) (x : ℝ) : ℝ :=
2   ∑ n ∈ Finset.range (N + 1), a n * Real.cos (2 * (n : ℝ) * x)
3
4 lemma lobachevsky_cosinePolynomial (N : ℕ) (a : ℕ → ℝ) :
5   (∫ x in Set.Ioi 0, (Real.sinc x) ^ 2 * cosinePolynomial N a x) =
6   ∫ x in (0 : ℝ)..Real.pi / 2, cosinePolynomial N a x
```

4.3 Uniform approximation on the additive circle

It remains to approximate every continuous function satisfying the hypotheses of Theorem 1 by cosine polynomials. More precisely, for every $\varepsilon > 0$, we seek

$$p(x) = \sum_{n=0}^N a_n \cos(2nx)$$

such that

$$|f(x) - p(x)| < \varepsilon \quad \text{for all } x \in \mathbb{R}.$$

Periodicity allows f to descend to a continuous function on

$$\text{AddCircle } \pi = \mathbb{R}/\pi\mathbb{Z}.$$

Mathlib's theorem `span_fourier_closure_eq_top`, from `Mathlib.Analysis.Fourier.AddCircle`, states that finite Fourier sums are dense among continuous complex-valued functions on this circle. In the formal proof, we denote this local density statement by `hS_dense`.

The reflection condition, together with periodicity, implies that f is even. Taking the real even part of a Fourier approximation and pairing the frequencies k and $-k$ removes the sine terms and produces the required cosine polynomial.

```
1 lemma exists_cosinePolynomial_uniform_approx_of_reflection {f : ℝ → ℝ} (
  hf_cont : Continuous f)
2   (hf_periodic : Function.Periodic f Real.pi)
3   (hf_reflection : ∀ x : ℝ, f (Real.pi - x) = f x) {ε : ℝ} (hε : 0 < ε) :
4   ∃ N : ℕ, ∃ a : ℕ → ℝ, ∀ x : ℝ, |f x - cosinePolynomial N a x| < ε
```

4.4 Passage to the uniform limit and the formal theorem

Define

$$L(g) = \int_0^{+\infty} \text{sinc}^2(x) g(x) dx, \quad R(g) = \int_0^{\pi/2} g(x) dx.$$

We conclude by contradiction. Suppose that $L(f) \neq R(f)$, and set

$$\eta = \frac{|L(f) - R(f)|}{2\pi} > 0.$$

Choose a cosine polynomial p satisfying

$$|f(x) - p(x)| < \eta \quad \text{for every } x \in \mathbb{R}.$$

Uniform approximation gives

$$|L(f) - L(p)| \leq \frac{\pi}{2}\eta, \quad |R(p) - R(f)| \leq \frac{\pi}{2}\eta.$$

Since Equation (10) gives $L(p) = R(p)$, we obtain

$$\begin{aligned} |L(f) - R(f)| &\leq |L(f) - L(p)| + |R(p) - R(f)| \\ &\leq \pi\eta = \frac{1}{2}|L(f) - R(f)|, \end{aligned}$$

a contradiction. Thus $L(f) = R(f)$.

In Lean, the two estimates are supplied by `abs_integral_sinc_sq_mul_sub_le` and `abs_intervalIntegral_sub_le`. Together with the cosine approximation theorem, they yield the final result.

```
1 theorem lobachevsky_integral_formula {f : ℝ → ℝ} (hf_cont : Continuous f)
2   (hf_periodic : Function.Periodic f Real.pi)
3   (hf_reflection : ∀ x : ℝ, f (Real.pi - x) = f x) :
4   (∫ x in Set.Ioi 0, (Real.sinc x) ^ 2 * f x) = ∫ x in (0 : ℝ)..Real.pi / 2,
  f x
```

5 New Directions and Conclusion

In this paper, we have formalized the Dirichlet integral as the limit of its truncated integrals, without identifying it with a Lebesgue integral of the non-integrable sinc function. The detour through sinc^2 provides both a proof of the Dirichlet integral and an independent evaluation of an absolutely convergent integral. From these two results, we derived the Dirichlet cutoff and several quadratic and bilinear trigonometric identities. We then used the same analytic foundation, together with Fourier approximation on $\text{AddCircle } \pi$, to obtain Lobachevsky's integral formula.

The present development suggests several natural extensions. At the level of integration theory, the Dirichlet integral illustrates the need for a convenient formal treatment of conditionally convergent improper integrals. In this paper, such an integral is expressed directly as a limit of integrals over bounded intervals. A more systematic interface for improper integrals could collect the usual operations (changes of variables, integration by parts, and comparison of different truncations) and make similar formalizations more direct.

The results on the Dirichlet cutoff also point toward further applications in Fourier analysis. Its convergence to the Heaviside function is a basic example of the way oscillatory kernels recover discontinuous functions. A natural continuation would be to use these results in formal proofs of Fourier inversion and of convergence theorems at points of discontinuity.

The most immediate extension of the Lobachevsky part concerns higher even powers of the sinc function. Jolany developed formulas for integrals involving sinc^{2n} and symmetric periodic weights [14]. Already for the fourth power, the result takes the form

$$\int_0^{+\infty} \text{sinc}^4(x) f(x) dx = \int_0^{\pi/2} f(x) dx - \frac{2}{3} \int_0^{\pi/2} \sin^2(x) f(x) dx.$$

The difference with the squared sinc formula is significant. For sinc^2 , only the constant cosine mode contributes to the integral, whereas higher powers retain several low-frequency modes. Formalizing these extensions would therefore require a broader collection of mode identities, but the approximation on $\text{AddCircle } \pi$ developed here should continue to provide the passage from cosine polynomials to continuous periodic functions.

Finally, the Dirichlet integral provides an elegant real-analytic route to Laplace inversion that avoids the residue theorem, and more generally supports the development of the theory of Laplace transforms. This approach is used in the companion formalization [16].

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